AI Engineering Learning Plan

A structured 12-week roadmap based on 12 top open-source GitHub repositories (combined > 1.05 million stars)

This plan organizes the repositories from the popular LinkedIn post into a progressive learning path. Work through the phases in order. Each phase builds on the previous one. Clone the repos, run the notebooks, and complete at least one hands-on project per phase.

Master Learning Roadmap (12 Weeks)

Phase / Week	Focus Area	Primary Repositories	Key Goals & Activities	Est. Time
Phase 1 Weeks 1–3 Foundations	Math, ML basics & LLMs from scratch	1. rohitg00/ai-engineering-from-scratch 2. rasbt/LLMs-from-scratch 3. karpathy/nanochat	• Complete early phases of AI Engineering from Scratch • Implement GPT-like model step-by-step • Train a small ChatGPT-style model end-to-end • Understand tokenization, attention, pretraining	25–35 hrs
Phase 2 Weeks 4–6 Core Courses	Structured LLM learning & Generative AI apps	4. mlabonne/llm-course 5. microsoft/generative-ai-for-beginners	• Follow LLM Fundamentals + Scientist tracks • Complete Microsoft 21-lesson course • Build text generation, chat & basic RAG apps • Practice with Python & TypeScript examples	20–30 hrs
Phase 3 Weeks 7–9 Agents & RAG	AI Agents and advanced Retrieval systems	6. microsoft/ai-agents-for-beginners 7. NirDiamant/RAG_Techniques 9. patchy631/ai-engineering-hub	• Master agent design patterns & tool use • Implement 8–10 advanced RAG techniques • Build multi-agent systems & agentic RAG • Study production tutorials from AI Eng Hub	25–35 hrs
Phase 4 Weeks 10–12 Production & Modern Stack	Apps, MCP, Skills & Real-world systems	8. Shubhamsaboo/awesome-llm-apps 10. anthropics/claude-cookbooks 11. modelcontextprotocol/servers 12. anthropics/skills	• Clone & customize 4–6 ready LLM apps • Study Claude agent patterns & tool use • Build/use MCP servers for tool access • Create custom Agent Skills • Deploy at least 2 portfolio projects	25–35 hrs

Detailed Phase Breakdown

Phase 1 – Foundations (Weeks 1–3)

Priority	Repository	What to Do
Start here	rohitg00/ai-engineering-from-scratch	Follow phases 0–6 (setup, math, ML, DL, transformers, LLMs). Build every algorithm from scratch.
Core deep dive	rasbt/LLMs-from-scratch	Complete chapters 2–7: data, attention, GPT model, pretraining, classification & instruction fine-tuning.
End-to-end practice	karpathy/nanochat	Run the speedrun script. Train a small model and interact with the ChatGPT-like UI.

Phase 2 – Core Courses (Weeks 4–6)

Priority	Repository	What to Do
Roadmap + notebooks	mlabonne/llm-course	Complete LLM Fundamentals (if needed) then The LLM Scientist and Engineer tracks. Use Colab notebooks.
Structured lessons	microsoft/generative-ai-for-beginners	Work through all 21 lessons. Build apps with Python/TypeScript using Azure/OpenAI APIs.

Phase 3 – Agents & Advanced RAG (Weeks 7–9)

Priority	Repository	What to Do
Agent fundamentals	microsoft/ai-agents-for-beginners	Complete the 18 lessons on design patterns, tool use, agentic RAG, multi-agent, and deployment.
RAG mastery	NirDiamant/RAG_Techniques	Implement at least 10 techniques from the 40+ notebooks (HyDE, fusion, GraphRAG, evaluation, etc.).
Production projects	patchy631/ai-engineering-hub	Pick 5–8 tutorials matching your interests (RAG, agents, fine-tuning, eval, MCP).

Phase 4 – Production & Modern Stack (Weeks 10–12)

Priority	Repository	What to Do
Build & ship apps	Shubhamsaboo/awesome-llm-apps	Clone, run, and customize 4–6 agents/RAG apps. Add them to your portfolio.
High-quality patterns	anthropics/claude-cookbooks	Study agent workflows (chaining, routing, orchestrator-workers, evaluator-optimizer) and tool use.
Tool protocols	modelcontextprotocol/servers	Explore reference MCP servers (filesystem, git, memory, fetch). Build a simple custom server.
Agent Skills	anthropics/skills	Study the SKILL.md pattern. Create 1–2 custom skills for your own use cases.

Practical Tips for Success

- **1. Hands-on first:** Always clone the repo and run the code. Reading alone is not enough.
- **2. Portfolio projects:** By the end of each phase, publish at least one project on GitHub with a clear README.
- **3. Compute:** Use Google Colab, Kaggle, or affordable cloud GPU spots (RunPod, Vast.ai) for training experiments.
- **4. Time management:** Aim for 8–12 hours per week. Consistency beats intensity.
- **5. Prerequisites:** Solid Python, basic linear algebra, and Git. Strengthen math via Phase 1 materials if needed.
- **6. Community:** Star the repos, open issues when stuck, and join related Discords / LinkedIn discussions.
- **7. Iteration:** After finishing the 12 weeks, revisit earlier repos with deeper understanding and refine your projects.

All repositories are free and open-source. This plan is a suggested path — adjust the pace and depth based on your background and goals.
Generated from the 12 open-source GitHub repositories highlighted for AI Engineers.